Ibrahim EL KASSIMI

ibrahim.el_kassimi@ens-paris-saclay.fr | linkedin.com/in/ibrahim-e-724431245 | github.com/IBRAHIM-EL-KASSIMI

Education

B.S. and M.S. in Electrical Engineering (EECS) September 2024 – Expected September 2028
ENS Paris-Saclay *Gif-sur-Yvette, France*

- Relevant coursework: Image Processing, Deep Learning, Diffusion Models, Image Priors, Bayesian Methods, Reinforcement Learning, Inverse Problems, Numerical Analysis and Optimization, Control Systems, Embedded Systems

Research Experience

Research Intern October 2026 – Expected August 2027
Priors for Remote-sensing Image Super-resolution Models

Research Intern: Event-Based Vision and GNNs | PyTorch Geometric April – Aug 2026
LIRIS, Imagine team *Lyon, France*

- Proposed **density-adaptive neighborhoods**, replacing the fixed-radius graph and reducing mean graph degree by **73%** (24.9 to 6.7) while improving classification accuracy to **88.2%** on N-CARS.
- Identified a key limitation in event-stream modeling: fixed-radius graphs waste edges in sparse regions and miss structure in dense ones; evaluated **7 adaptive-neighborhood configurations** to address this inefficiency.
- Evaluated 9 time-rescaling values ($\beta = 5 \times 10^{-7}$–5000), finding that $\beta = 800$ reduced mean graph degree by **81%** (24.89 to 4.69) while maintaining **87.78%** classification accuracy.

Experience

Scientific Data Processing | Python, Pandas, PyQt Feb 2025 – May 2025
Kimialys (Biosensing startup) *Paris, Remote*

- Reduced a previously manual workflow from **3–4 hours** to only **a few minutes**, improving efficiency, consistency, and traceability in experimental analysis.
- Built a **Python-based data processing pipeline** for 3 assay cycles (1,025 time points across 21 measurement channels), detecting and imputing missing values and standardizing molecular reflectivity data for analysis.

Projects

Autonomous Airship | *Python, OpenCV, Raspberry Pi, Sensor Fusion* 2025

- Led the vision team in designing and integrating the **sensor and vision subsystem** for an autonomous airship, combining a camera, AprilTag-based localization, IMU, and distance sensing to estimate position and altitude in real time, with position estimation error of at most **10 cm**.
- Developed a **computer vision pipeline** for visual detection and pose estimation, enabling a **20-meter autonomous mission** to approach a target, pick up a container with a programmed gripper, and deposit it at a second target; my team won the challenge.

Technical Skills

Languages: Python, C++, MATLAB, SQL
ML / CV: Machine Learning, Deep Learning, Computer Vision, CNNs, PyTorch, TensorFlow, scikit-learn, OpenCV, PyTorch Geometric, NumPy, Pandas
Methods: Algorithms, Mathematics, Graph Neural Networks, Graph Construction, Data Analysis, Technical Writing
Tools: Git, GitHub, LaTeX, Raspberry Pi, Embedded Systems, Linux, MATLAB/Simulink, AWS

Publications

"Réalisation d'un dirigeable autonome - CultureSciences de l'Ingénieur." *La Revue 3EI*, 117 (2025)

Leadership

Ensurance AI Assistant Hack'In Saclay *(2nd / 11 teams)*

- Built an AI-powered insurance contract Q&A assistant and ranked **2nd / 11 teams** in Hack'In Saclay.

Interests

Rollerblading, football